

STRATEGIC RESEARCH SUMMARY · TEAM 100 · 2026-06

The Operating System for the Small Farm

מערכת ההפעלה של החווה הקטנה

Vision Re-Lock · Platform Decision · Competitive Intelligence · Schema Reference

SmallFarmsAgents · Hebrew-first · headless over farmOS

From open question to a locked, evidence-backed direction

1

Vision Re-Lock

North star, 5 pillars, audience, Freemium model — approved in-session.

2

Platform Decision

farmOS chosen (vs LiteFarm), headless, 3-tier delivery.

3

Competitive Intelligence

12 competitors + 2 OSS, mapped across internal + 4 AI engines.

4

Tend Reference

Field-exact schema from real farm exports → build-ready blueprint.

12+2

competitors + OSS benchmarks

4

external AI research engines

16+

research sub-agents run

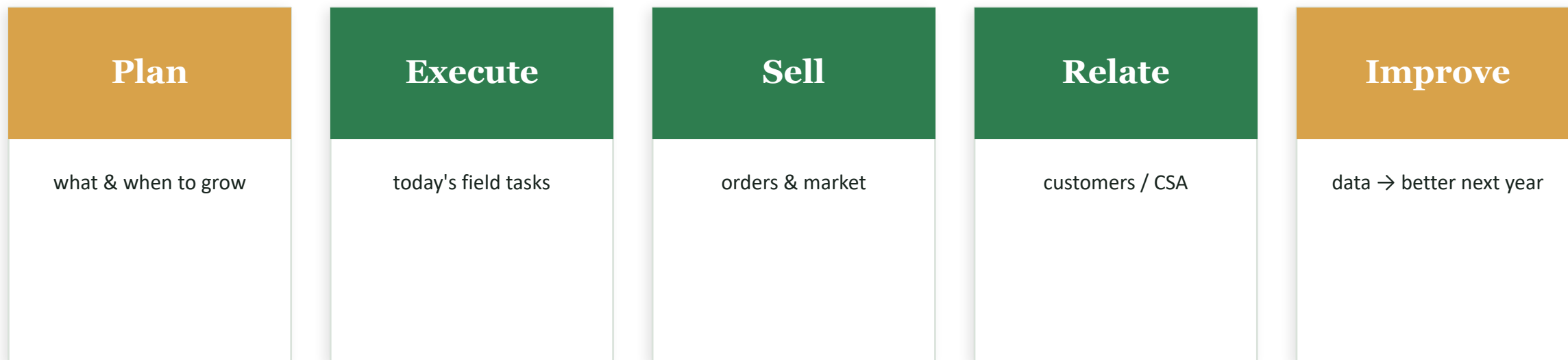
100%

findings converge

The page every grower opens in the morning

מערכת ההפעלה של החווה הקטנה

End-to-end across five pillars — unified in one role-aware daily cockpit (Manager + Worker).



Brain (Plan / Improve) = our agronomic-economic engine · Operations (Execute / Sell / Relate) = farmOS + integrations

Two audiences, one funnel

Commercial market gardener

- Paying core (JM Fortier / bio-intensive style)
- It's a business → real willingness to pay for tools that make/save money
- What the notebookLM spec & the data engine target

Home / private grower

- Brand engine, not just free users — Nimrod's course customers
- Free community → trust → courses, consulting, custom dev
- Establishes the brand for SFA and for Nimrod as consultant

Monetization: Freemium + strong indirect. Free Hebrew Crop-Book + calculators = the brand front (build first, build excellently). Paid = durable farm management + economics. Indirect revenue (brand → courses → consulting) may exceed direct.

farmOS — adopted headless

1

API built for integration

Documented JSON:API + OAuth2 + official Python client. (LiteFarm's API is internal-only → brittle.)

2

Flexible data model

Asset / Log / Quantity / Term + custom fields hold our 13-topic taxonomy & provenance. (LiteFarm schema rigid.)

3

Blessed, documented self-host

Docker — fits our stack. (LiteFarm self-host unsupported.)

Why headless?

- We build our own Hebrew/RTL UI — solves UX + Hebrew + GPL boundary at once.
- License GPL-2.0 (not AGPL): SaaS + headless → our engine stays proprietary.
- Bus-factor managed: we hold the code, data & frontend; Farmier = commercial backstop.

LiteFarm's deciding negatives: zero RTL/Hebrew + internal-only API → headless neutralizes its UI advantage.

Three tiers — cost scales with commitment

No account

Stateless

- Crop Book + 14 calculators
- Plan + export, anonymous
- ∞ concurrency · ~0 cost

FREE

Rented sandbox

Ephemeral instance

- Full farmOS, spun up on demand
- Bundle loaded → saved → reset
- Fits farming's seasonality

PAY-PER-USE

Permanent instance

Always-on

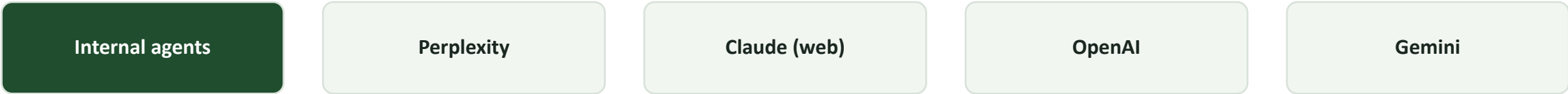
- For serious operators
- Live volume, always mounted
- Full operational record-keeping

SUBSCRIPTION

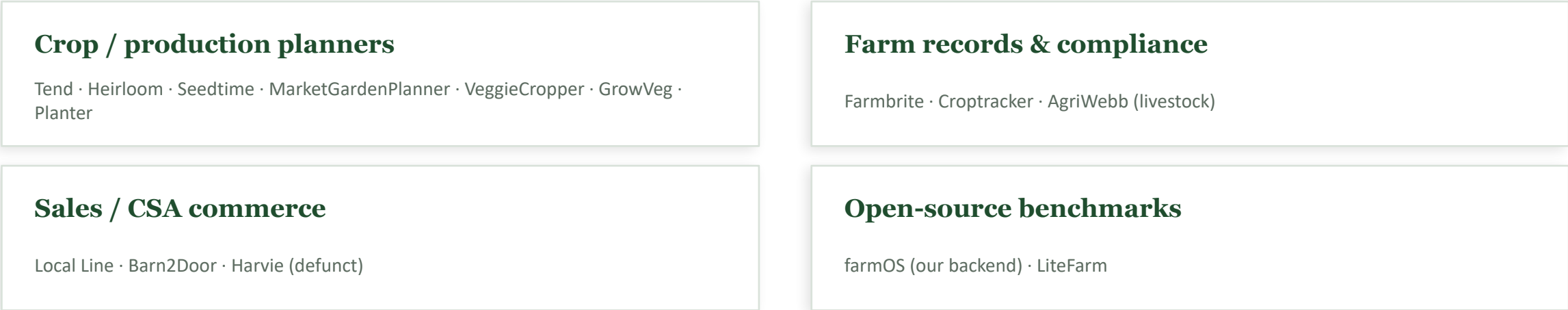
Unified by the "bundle" = a DB/volume snapshot. Promotion ladder: draft → save bundle → keep always-on. Ephemerality kills the upgrade-debt of always-on instances.

Quadruple-source competitive intelligence

Every finding cross-checked across internal research agents and four independent AI engines.



The market splits into four shallow, un-moated clusters — none spans the whole farm:



Three open lanes nobody serves

1

No one closes the market-price → plan → profit loop

All economics is retrospective or self-entered. Heirloom lists it on its public roadmap — unshipped. → Our #1 wedge.

2

Zero Hebrew / zero RTL — anywhere

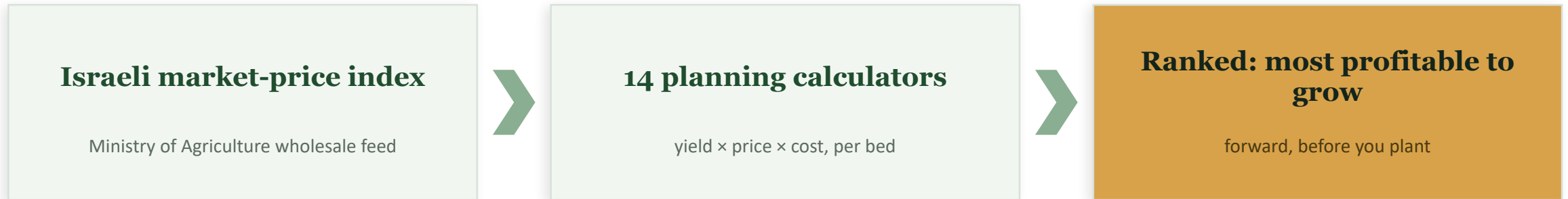
Even \$64M AgriWebb & JM Fortier's Heirloom built no RTL. The only Israeli product is a printed paper calendar. → Our moat.

3

The production ↔ sales loop is industry-wide open

Sales tools never feed planning; planners never pull demand. → Our unified 5-pillar OS is unbuilt anywhere.

"What is most profitable to grow — right now?"



Nobody else does this. Every competitor that touches money tracks it *backward* (your own books) or uses self-entered prices. SFA turns the record-book into a **forward economic engine** — agronomy × live Israeli market prices.

Hebrew-first, RTL-native, Israeli climate & market

Structural, not cosmetic. RTL must be built in from day one — every field, calendar, bed-map and notification.

0

competitors with any Hebrew or RTL — out of 12 + 2 OSS

1

Israeli home-garden product exists — a printed paper calendar

\$64M

AgriWebb funding — still built no RTL (nor full Spanish)

Plus Israeli specifics no global tool will build: Negev vs Galilee seasonality, local pests, holidays, Shabbat-friendly views, water-salinity.

Where (almost) nobody plays

1

Market-price → profit planning

The forward economic engine. Universal gap.

2

Hebrew / RTL / Israel

Uncontested. A defensible moat, not a feature.

3

The unified morning cockpit

Plan+Execute+Sell+Relate in one role-aware screen.

4

Beautiful mobile canvas + cockpit

Planter owns mobile; Seedtime owns the cockpit; nobody owns both.

What we adopt — three layers

Tend — the spine

Crop → Growing-Template → Planting → Task → Harvest → Lot → Inventory. Multiple templates per crop, cascade/clone. Maps 1:1 to farmOS Asset/Log/Quantity/Term.

LiteFarm — crop depth

Deepest OSS crop schema (crop type → varietal → management plan, estimated yield + value/yield). Second reference.

farmOS — the substrate

Asset/Log/Quantity/Term + custom fields. The operational ledger; we own the opinionated UI on top.

Field-exact, validated: 4 years of real Tend exports from Nimrod's own farm are already in-repo (64-column CROP_PLAN, metric, ) + an existing importer. Inference → fact.

Match the table stakes — beat on what matters

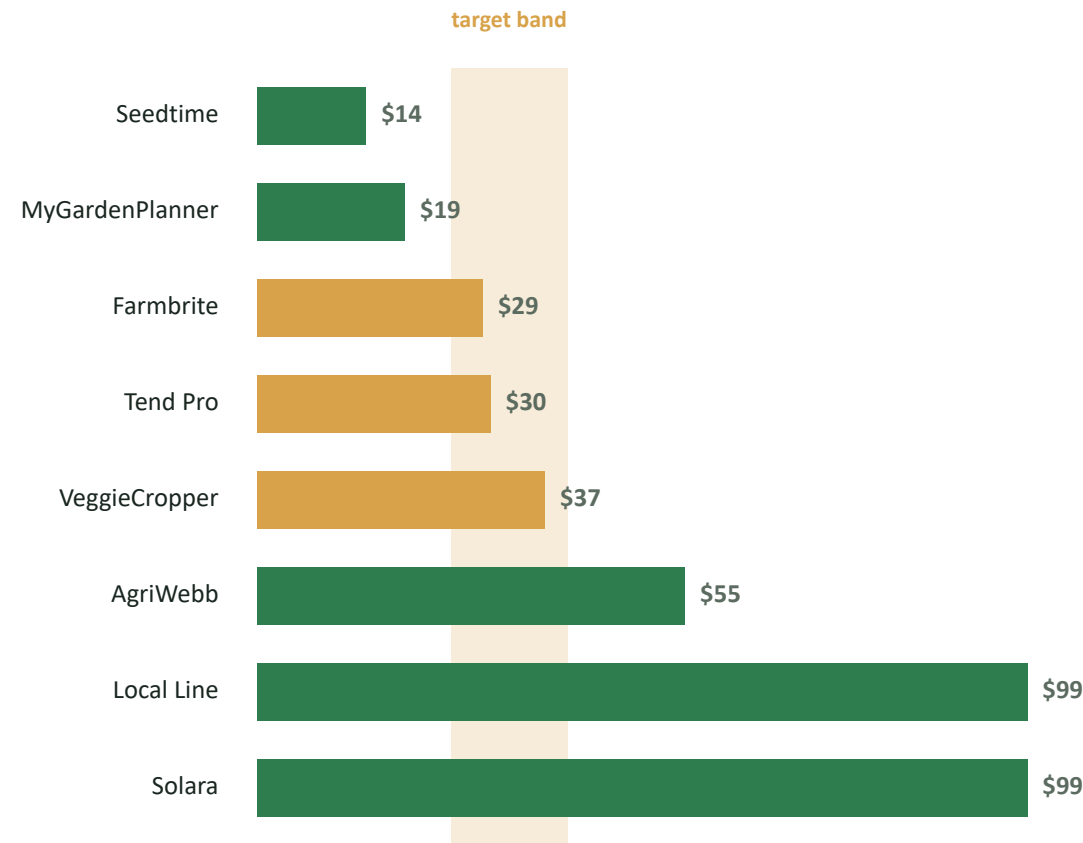
MATCH (table stakes)

- Backward-from-harvest scheduling + bed-turnover
- Connected loop + public order forms + availability sheets
- Crew / manager accounts with task autonomy
- Succession + rotation as the planning hook
- CSA box-first backward planning
- The 14 calculators (seed/yield/spacing/succession)

BEAT (where we win)

- **Market-price → forward profit loop (the hero)**
- **Hebrew-first / RTL + Israeli climate & market**
- Curated agronomic KB + 14 calculators (rivals: free fields, no science)
- Openness & durability (farmOS + export) vs closed, bus-factor-1 rivals
- Commercial tier = capability (Sell+Relate+finance), not "more beds"

Where a serious tool sits — and our target



SFA target

~\$25–40 / mo

equiv.  — below Tend Pro, justified by the economics engine

- Free tier = table stakes (rare → our advantage).
- Croptracker ~\$275/mo (enterprise, off-scale).
- Flat, transparent, no setup fees, no % of sales.

Monthly \$ equivalent · Croptracker ~\$275 (enterprise, off-scale)

From Nimrod's years inside Tend

① The harvest loop is priority #1

Harvest-list → field-report → accumulated learning is the most valuable AND the most painful loop. The crew never understood what to update. Make it fast, structured, worker-obvious — the gold lives here.

② Speed is a hard requirement

Many Tend operations were simply slow. Slowness kills field adoption. Treat responsiveness as a non-negotiable NFR.

③ Keep what worked

Time+space colour bed-map, auto-generated tasks, auto harvest-lists, clone-crop-shows-on-map.

④ Redo what didn't

Everything must run much faster; all field reporting must be far simpler & more structured for the worker.

Research is complete → proceed to Phase 0

12 competitors + 2 OSS · 4 AI engines · internal agents — all converge. No further CI rounds needed.

1

Data model

farmOS Asset/Log/Quantity/Term × Tend spine × 13-topic taxonomy × 14 calculators

2

3-tier delivery

No-account · rented sandbox · permanent — the bundle lifecycle

3

The morning cockpit

Role-aware (Manager/Worker), RTL-native, harvest-loop-first, fast

4

The wedge, wired

Ministry-of-Ag price index → calculators → 'most profitable to grow'

Open decisions carried forward: D1 pricing specifics · D2 multi-tenancy at scale · D3 Sell/Relate = build (no clean integration target).

מערכת ההפעלה של החווה הקטנה

The operating system for the small farm.

Wedge + moat confirmed against 12 competitors + 2 open-source benchmarks.

Next: Phase 0 — technical design